City Ditty User Guide

Table of Contents

City Ditty User Guide 1

Overview 2

A short primer on Soundscapes and Soundscape design 2

Research Outputs 2

City Ditty location on your PC 3

Directory structure 3

The importance of User ID / Participant # 4

Audio Playback 5

Three Play modes and Data Collection (Overview) 6

Free Design mode 6

Sound-Awareness Session mode 7

Experiment mode 8

City Ditty Controls 12

Desktop controls - Movement 12

Desktop controls – Object Placement 13

HMD VR controllers 13

HMD VR controls – Object placement 15

HMD VR controls – Window placement 15

Further Documentation 16

Overview

This documentation briefly introduces the software's three modes of operation (***Free Design Mode***, ***Interactive Modules Mode***, and ***Experiment Mode***) and provides technical details on several required functions. This includes where files are saved on your PC, controls for both Desktop and HMD versions, and provides an overview of the other documents that provide more detail on how to analyze any data that City Ditty produces.

All data is collected locally on your PC and is not shared elsewhere.

- **Note:** On the software's first load up, windows might ask you if you want to give it permission to access the internet. You can say yes or no. It doesn't matter as City Ditty does not need anything online or attempt to connect anywhere online.
    - It does, however, have a button that will open information pages in your browser. This will work regardless of permissions, as it is just a simple hyperlink that is sent to your browser.

This software was designed and created by Richard Yanaky as a part of his doctoral studies at McGill University. An overview of the City Ditty project, supporting funding agencies, collaborators, etc. can be found here.

A short primer on Soundscapes and Soundscape design

1) Be sure to read this short webpage on Why Sound Matters. It illustrates 10 simple principles of soundscapes to help you understand both the importance of sound, and how we can use it as a resource to support healthier and happier cities!

2) There is a built-in Sound-Awareness session within this software that introduces users to both how to use the software, as well as several concepts of soundscape design. Give it a try!

3) The Sounds in the City website also showcases other research projects and case studies to help learn more about how to use sound as a resource.

Research Outputs

For updated research-related outputs, see Richard's ResearchGate (as Developer of City Ditty) or the different Sounds in the City Project pages for related soundscape outputs.

City Ditty location on your PC

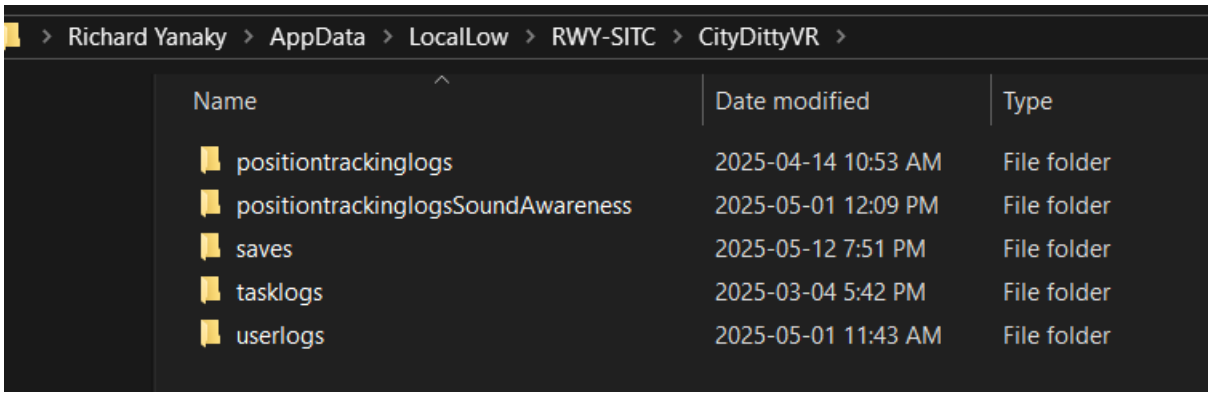
The City Ditty software itself can be stored anywhere on your PC and does not require installation. It only runs on windows (for now) and contains all software generated files (e.g. save files, experiment files) in the window user account's AppData folder. By default, AppData is stored in the user's profile in their AppData\LocalLow folder*. All files will be stored here, regardless of where you store the City Ditty software.

e.g., C:\Users\Richard Yanaky\AppData\LocalLow\RWY-SITC\CityDittyVR\

**Why this awkward location? Because it was the first place I was able to reliably save and load files from during early development, without learning more ins and outs of interacting with the operating system. I recommend creating a desktop shortcut to this location for ease of access.*

Directory structure

The following directory structure is used. Folders and files will be generated automatically as needed.



Positiontrackinglogs

- Contains position tracking logs from Experiment Mode.
- See **PositionTrackingLogs-ExperimentMode.pdf** for details on analyzing the logs.

PositiontrackinglogsSoundAwareness ***(to be renamed later)***

- Contains position tracking logs from Interactive Modules Mode.
- *Interactive Modules* was originally called *Sound Awareness Sessions*. I'm in the process of renaming it in multiple locations.
- See **PositionTrackingLogs-InteractiveModules.pdf** for details on analyzing the logs.

Saves

- Contains all saved soundscape designs (called saved scenes), including sample scenes, user created scenes, and scenes for experiments.
- Sample Scenes are automatically created on software run. They cannot be overwritten in City Ditty. If overwritten or deleted in the file explorer, they will be recreated on next software startup.
- Saved scenes are saved as .JSON files which contains all data required to recreate a scene.

    - This includes all XYZ positions and other related data for all sound sources. These can be extracted if you want to use them for other purposes.
- All design files, other than Sample Scenes can be freely renamed.
- For a description of how to setup Experiment files, see the section on ***Experiment Mode***, below.

Tasklogs

- Includes tasklogs from Interactive Modules Mode.
- This is the 1-5 Likert scale data that asks users to rate individual Tasks.
- See **TaskLog (Likert Data) Descriptions.pdf** for details on analyzing these logs.

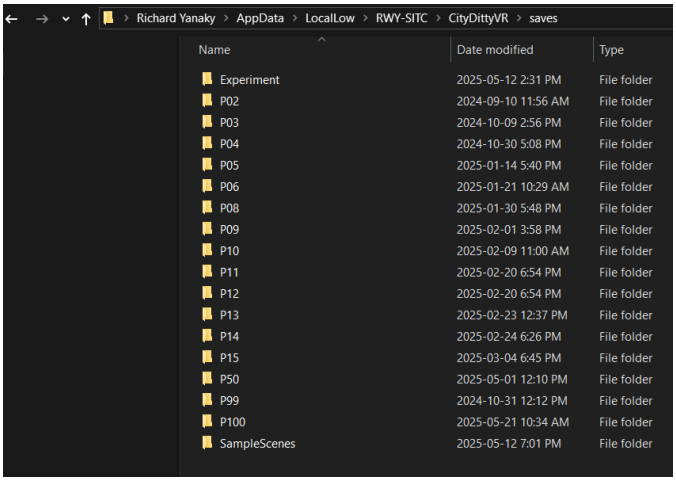
Userlogs

- Includes Userlogs from Interactive Modules Mode.
- This are the logs of all interactions taken by the user for usability tests and other related purposes. E.g., what they clicked on, settings changed, etc.
- See **Userlog descriptions.pdf** for details on analyzing these logs.

The importance of User ID / Participant

Whether you are using Free Design, interactive Modules, or Experiment mode, it is always being operated with a User ID or Participant # (same thing. It is referred to a friendlier 'User ID' name in software, but participant # during analysis and directory structure). This is used in all generated data logs, as well as used as an identifier to give separate workspaces to users to keep their user-generated saved scenes separate. It also saves their logs/saved scenes in separate folders.

All of the above folders use the same subfolder structure, with any logs or saved scenes being held in a **Pxx** folder (for Participant #). The saves folder also contains the SampleScenes folder and the Experiment folder, which contains any experiment settings.



Audio Playback

The current version of City Ditty only supports audio playback in Binaural. This means it is optimized for playback via headphones.

Later versions may expand this back out into simple stereo and complex surround sound (e.g. Ambisonics), but Richard is otherwise writing his dissertation at the time of writing this.

Three Play modes and Data Collection (Overview)

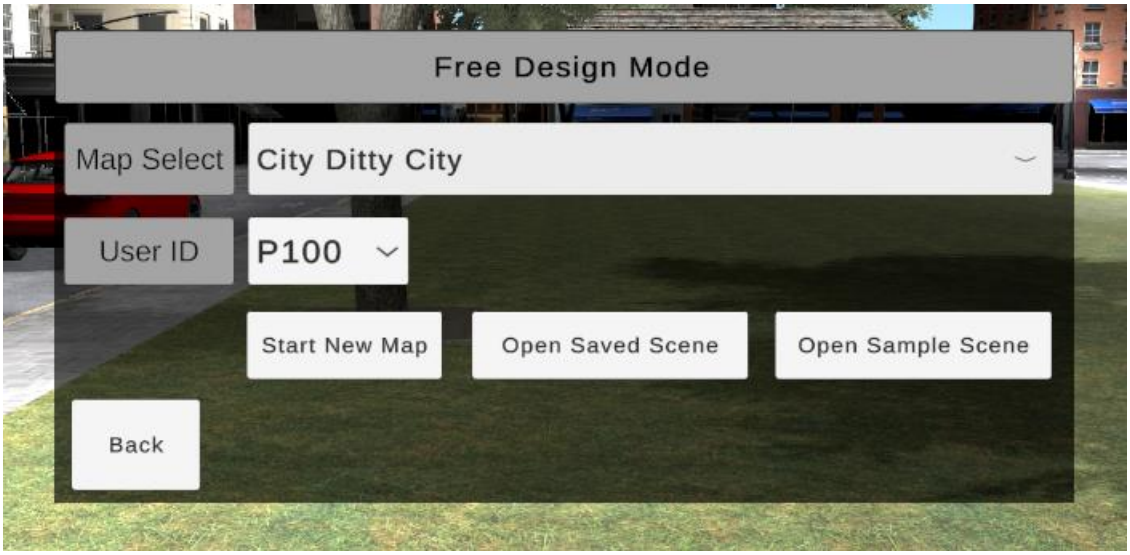
Three play modes currently exist.

Free Design mode

This is the standard mode for design. It gives users full access to all functionalities, with no data or metrics collected. To start, set the user ID, and then either open a saved scene or start a new empty map. There are currently two maps available: the City Ditty City (demo city), and Parque O'Higgins (Santiago, Chile).

The addition of additional sound sources, 3D objects, and maps is possible, but still requires developer collaboration, due to its complexity.

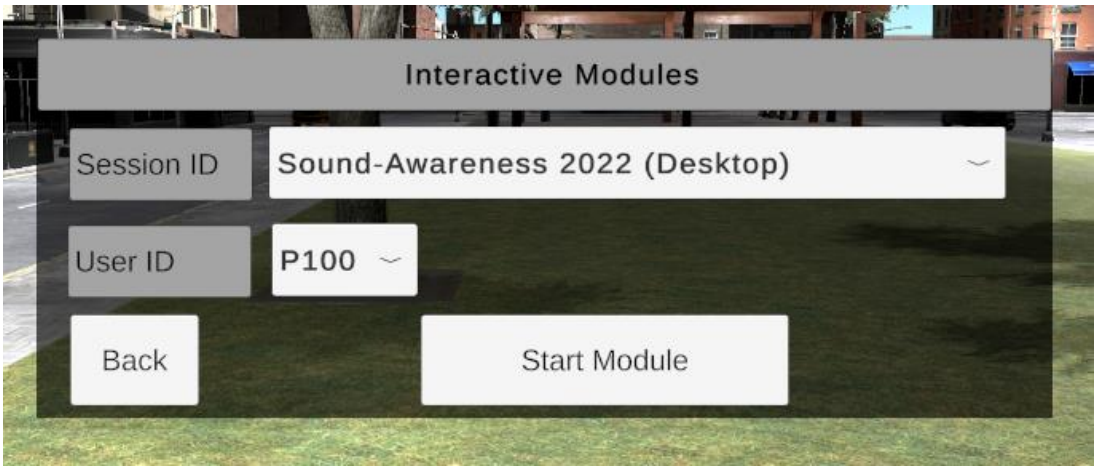
- See **Unity and Wwise contribution guide.pdf** for details on providing your own sound sources, 3D objects, and maps. Then contact the developer.



Interactive Modules mode

This guides the user through a list of Tasks, each with a set of instructions for the user to complete. This is currently used to guide the user through how to use the software and introduce them to different soundscape design methods to raise their sound-awareness. There are several forms of data collected here, including Likert scale questions at the end of each task, user-action logs, and position tracking.

Upon completion of a task, the software can ask 5-point Likert scale questions. The user has full access to all functionalities in this mode. It also records every instance of the user clicking on any button, sound source or other interactable object (user-action logs), as well as records the XYZ coordinate position of the user every 0.5 seconds (with adjustable intervals) to track their path through the virtual environment.



Built-in tutorial for City Ditty

The first designed Interactive Module was called *Sound-Awareness session,* which can be used as a tutorial for how the controls work, as well as introduce the user to several concepts in soundscape design. An older version was used as part of the software's user-centered design process, and has been discussed in the first major publication on City Ditty. Its HMD VR version continues this trend with an upcoming publication (in prep).

Platform for a UX study

This mode has also been used in a UX study which takes advantage of its task-based structure to observe and analyse user experience and performance. Publication in prep.

Create your own Interactive Module

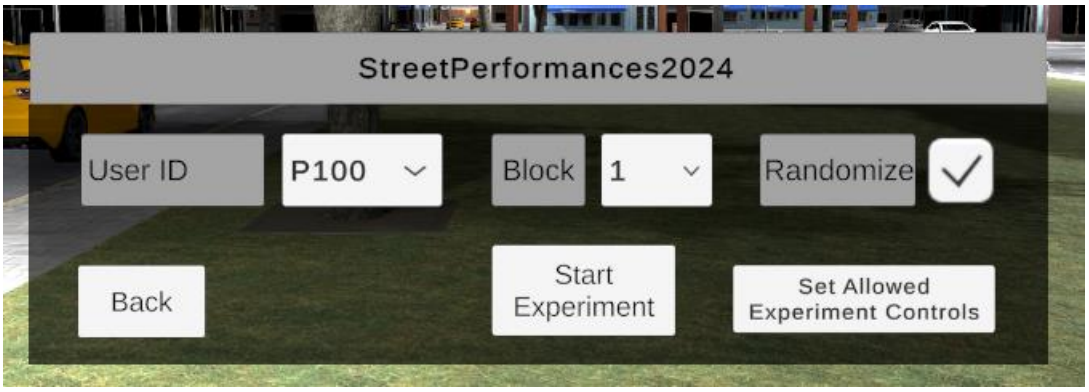
There isn't an interface to create your own yet – this needs to be done in collaboration with the developer still. Please contact Richard if interested.

If you only need people to walk around, listen, or evaluate soundscapes without changing them, consider using the Experiment Mode.

Experiment mode

This was created to help researchers study and evaluate the difference in people's behaviour in respect to different soundscape designs. The researcher can create any number of scenes required for their controlled study in Free Design mode, then load them into the deck for the participant to iterate through, either sequentially, or in randomized order. During this mode the user can only walk around the space and select final positions before moving onto the next one, with optional questionnaires between them. They cannot manipulate the soundscape in any way – only walk around, listen, and select final positions. This mode tracks the XYZ coordinate position of the user's position every 0.5 seconds (with adjustable intervals).

If manipulation of the soundscape by the participant is necessary, the Interactive Modules mode is recommended instead.

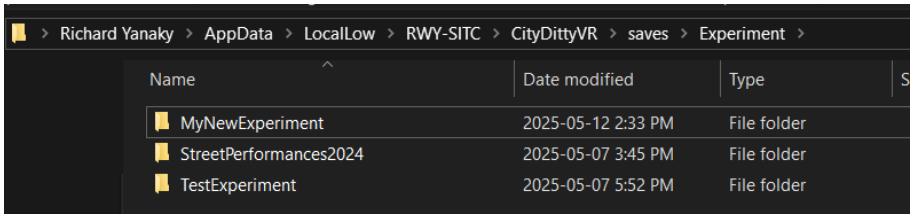


Creating your own experiment

If the researcher is using existing sound sources, 3D objects, and maps, the entire process can be completed from start to finish on their own using City Ditty. The first experiment of this manner has been used to explore the effect of music on the quality of urban public experience, and is in prep for a full publication which includes more details on methodology & analysis.

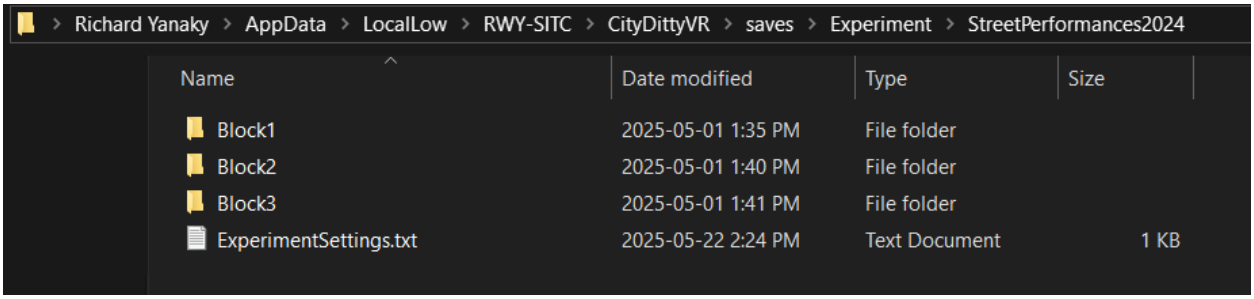
Experiment file structure follows the structure of Experiment Name → Block(s) → Scene(s). To setup your experiment, complete the following steps:

1) **Within the saves/Experiment folder, create a new folder and give it the name of your experiment.**
   - This example has three different experiments.



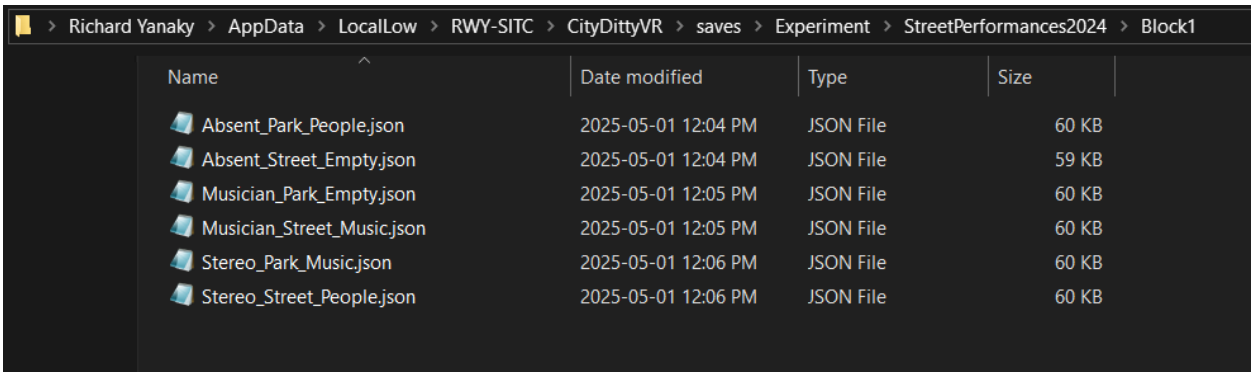
2) **Create *at least one* subfolder representing an experiment block**
   - You can create as many experiment blocks as you like.

- ***All folders in this directory are treated as blocks***.
  - *You can name folders whatever you want, but should follow a consistent alphanumeric order.* This is because all folders are assigned a block number.
  - If there are no subfolders, you will get an in-software error telling you to fix this.
  - This example of the Street Performances 2024 experiment has three blocks.
- An ExperimentSettings.txt file is automatically created on first run and is stored in this folder. Step 4 explains these settings.



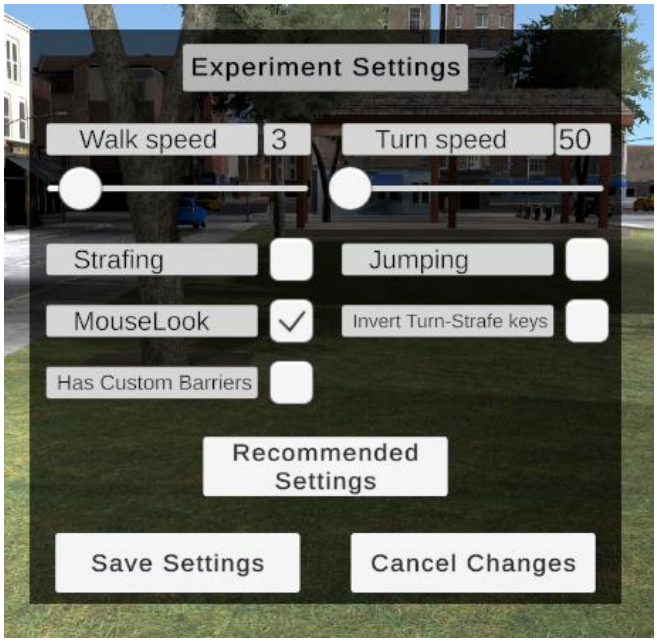
3) **Fill each of your blocks with the soundscape scenes that you created in Free Design mode.**

- Give your scene names descriptive filenames. These filenames are never visible to participants and will help you identify the scene in the generated datalogs.
- You can put in as many scenes as you like.
- If there are no scenes in a block, you will get an in-software error message, informing you to fix this.
- City Ditty does not check for issues in experiment design, beyond the technical. i.e., checking for the existence of blocks and scenes in them.
  - E.g., This street performance experiment has 6 scenes per block. City Ditty does not check for balanced conditions, for example. Experiment design remains the responsibility of the researcher.



4) **Assigning Experiment Settings**

- The experiment can already be run with default settings at this point! However, you can review and change the experiment settings before starting. This image is from the Desktop version:



- Most of these settings are in relation to user controls on the desktop. The default settings, which can be restored by pressing the Recommended Settings button, are set to support the simplest of controls, accessible to the widest variety of users, at a realistic normal walking speed.
  - User controls are described in the following ***City Ditty Controls*** section.
- **Has Custom Barriers** allows placing invisible barriers to prevent the participant from leaving the experiment designer's pre-determined experiment space. E.g., it prevents participants from exploring irrelevant side streets or other areas.
  - Currently, only the City Ditty City has this, as designed for the Effect of music on the quality of urban public experience experiment (image in link). These barriers currently enclose:
    - *The 4-block central park area*
    - *A 4-block sidestreet area*.
- These settings will be saved to the ExperimentSettings.txt file.
  - A warning is given upon trying to overwrite existing experiment values.
  - Values can also be changed in the ExperimentSettings.txt file itself, but is not recommended. I won't give support to any issues that arise when doing so.

5) Running the experiment

- Set the **UserID**, **Block #,** and hit start!
  - Randomization is enabled by default and should only be disabled with good reason.
  - Questionnaire is not provided. I recommend having a separate laptop with questionnaire available between scenes.
    - User tracking is disabled at questionnaire time between scenes.

- Don’t forget to include and match the **userID** and **scene #** (in the order they were presented in) on your questionnaire.
- The experiment logs keep track of the order in which the scenes were presented in. You will need to match these to the actual scenes presented, yourself, due to the randomization.
- Be sure to have a test run to ensure that your questionnaire has everything required to match up properly! Everything else required is included in the generated experiment logs.
- See **PositionTrackingLogs-ExperimentMode.pdf** for details on analyzing the logs.

City Ditty Controls

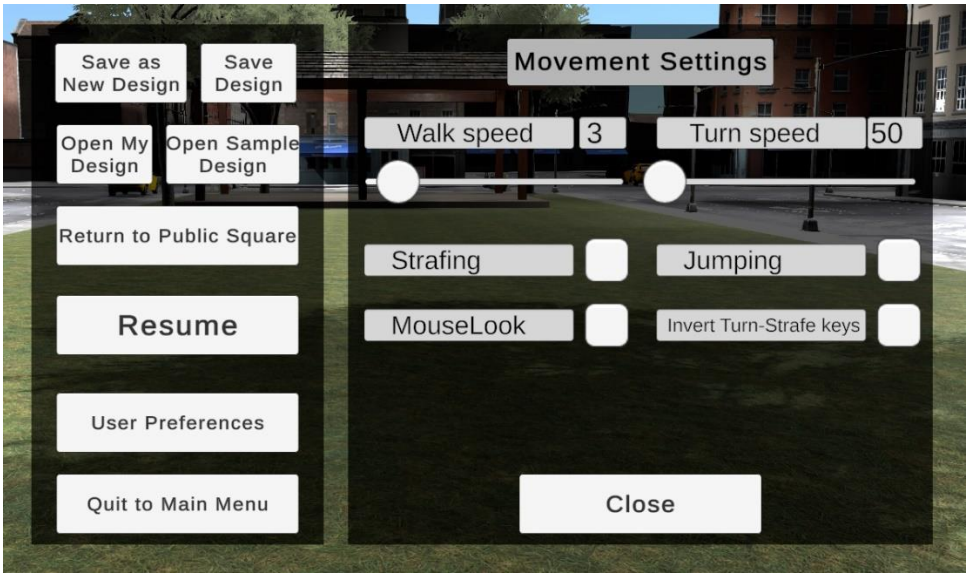
City Ditty operates in both Desktop PC with keyboard and mouse, as well as HMD VR – currently only tested with HTC Vive controllers.

Desktop controls - Movement

By default, the user can move around with the WASD and arrow keys on any standard keyboard. They are set to accommodate simple movements for a 3D environment, including walking forward, backwards, and rotate left and right, at a somewhat realistic walking speed and slow rotation to not disorient novice users. These settings can be changed, however.

In either the Free Design or Interactive Modules mode, the user can select Options - > User Preferences to load up the movement settings options. These can be used to adjust speeds and help more advanced users emulate more game-like controls for their preferences. These settings are shared between Free-Design and Interactive Modules modes and all participants. Personalized settings per user is planned for a later update.

Experiment Mode also allows these settings to be changed, but is stored separately on a per-experiment basis. It is highly recommended to keep them simple to accommodate realistic movement and accessibility to all users. More detail is provided in the Experiment Setup document.



Walk Speed

- A default walk speed of 3 represents a somewhat natural walking speed. This can be increased or decreased to suit user preference.

Turn Speed

- A default turning speed of 50 is used. This can be increased to suit user preference.

Strafing

- Strafing, or side-stepping is a common gamer movement, but is not enabled by default. Enabling this will use Q and E buttons to strafe.

Jumping

- Pressing the spacebar will allow the user to jump. This is disabled by default. It's mainly included for fun.

MouseLook

- When enabled, this will allow the user to press and hold the right mouse button to look up and down and rotate left and right. Pressing the middle mouse button will reset the looking direction to neutral.

Invert Turn-Strafe Keys

- This will swap turning keys with strafing keys to suit user preference. i.e. swap Q E with A D.

Shift Run Key

- Holding Shift while moving will increase movement speed until released.

Warning: This movement system currently uses a physics-based system that behaves a bit oddly at higher movement speeds. Likewise, jumping is a bit awkward and sometimes jumps quite high. It is not harmful to use yourself, but higher speeds are not recommended during demos or experiments. I will be replacing the movement system in the future which will address these issues.

Desktop controls – Object Placement

The control scheme changes for object placement. Once an object is selected for placement, stiffer controls for placement are utilized. These cannot be changed in User Preferences.

W S : remains forward and backwards.

A D : strictly moves left and right without turning (like strafing).

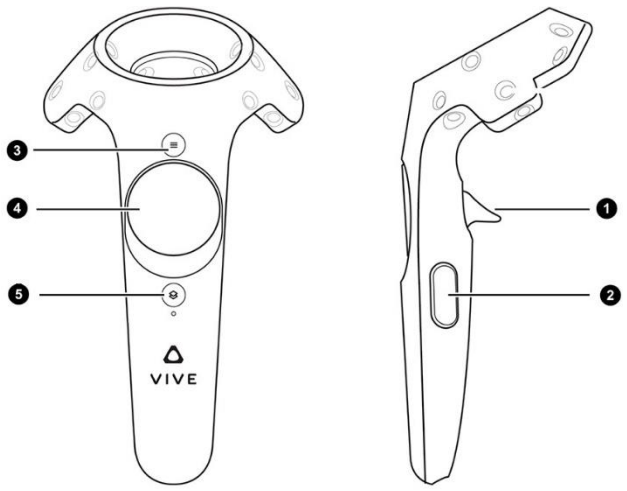
Q E : moves up and down.

Middle mouse button: Enables/disables object rotation with the mouse.

HMD VR controllers

The HMD version is designed to be a gentle experience. However, if you are known to be prone to vertigo, VR sickness, or any other conditions that may result in potential adverse VR side-effects, please consider using the Desktop version – it has all of the same functionalities. That said, all movements are designed to be natural to avoid any adverse VR side-effects as much as possible. I recommend trying it, unless you have known issues. Discontinue use if you experience discomfort.

City Ditty has only been tested to date with HTC Vive controllers. The actual HMD should not matter (we have run it on both a Vive Pro and Varjo X-3 headsets with no problem). The controls are listed below:



The Trigger buttons (1)

- This is the mouse button, basically. Point at things and select or click on them with the Trigger buttons. You'll also use this to click on any buttons or grab other menu items like sliders. This works with either controller.
    - Note: If you are touching the touchpad, the pointer will transform from straight to a curved pointer for teleportation. You cannot interact with items when in teleportation mode (described below).

The Grip buttons (2)

- They're on both sides of both controllers. When you want to grab, or pickup an interactable object to move it, you'll point your controller at the object, then grab it with a grip button.
- You drop objects by releasing the grip button.

The Menu button (3)

- Each controller has a menu button that will open up different windows.
- The LEFT-Hand Menu button opens a window that let's you interact with the soundscape.
    - This provides all menu items that are otherwise normally on screen already on the desktop version.
- The RIGHT-Hand Menu button opens and closes the in-software instructions that accompany the Sound-Awareness sessions.
- Both of these can be freely opened and closed.

The Touchpad (4)

- The virtual world is likely much bigger than wherever you're running the software, so you'll use this touchpad to teleport around. Either controller works for this.
    - Touching the touchpad with your thumb will transform your pointer to a curved teleportation pointer. Aim at the ground and press the top part of the touchpad to teleport there.

        - Note: You cannot click on menu items when in teleport mode. Raise your thumb up to transform the pointer back into a straight selector pointer.

System's Menu button (5)

- An additional button below the touchpad pauses the VR simulation entirely. We don't use it. If you accidentally hit it, press it again to resume play.

Note: There is a known rarely occurring bug that still exists where the menu and trigger controls stop behaving properly. Pressing the System's Menu button will fix/reset their behaviour. This will be fixed in a later version.

HMD VR controls – Object placement

Point at an interactable object, then press and hold a Side Grip Button on either controller to grab it. You can then move it around naturally.

- Don't stand too close! You will have better control over objects if you stand back a bit. Release the Side Grip button to place it.
    - This holds extra true for larger objects. Give yourself space!
- You can pull a held object closer or push it further away from you with long pulling and pushing movements.
- You can also hold an object with one hand while teleporting yourself and it to new locations with your other hand.
- To rotate an object, touch and hold left or right on the touchpad while holding the object.
- To create copies of an object, press the trigger button while holding the object.

HMD VR controls – Window placement

As you work in the 3D space, you will often need to move around. These windows are built to support this. Windows will always open directly in front of you, and will stay open, even when you move or look away.

- Windows can be moved by closing and re-opening them in the direction you're facing. You cannot grab and drag them to manually relocate them, as of yet.
- Important: Open windows in a clear space! Overlapping windows don't work well. They can also be blocked by other objects in this space. Be aware of your surroundings. This is a conceptual fact of working in 3D spaces that we need to adapt to.
- The Menu button will open and close windows.
- You can also point at sound sources and click on them directly with the trigger button to open a window to modify their respective settings.

Further Documentation

As indicated in the respective locations above, additional documentation can be found in separate documents. These include:

- **Position tracking logs for sound awareness mode:**
  - PositionTrackingLogs-InteractiveModules.pdf
- **Task questions for sound awareness mode**
  - TaskLog (Likert Data) Descriptions.pdf
- **Position tracking logs for experiment mode:**
  - PositionTrackingLogs-ExperimentMode.pdf
- **User-action logs for sound awareness mode (usability study data)**
  - Userlog descriptions.pdf
- **Unity and Wwise contribution guide**
  - Unity and Wwise contribution guide.pdf